

Gas Monitoring via an Integrated Stationary Sensor and Mobile Robot System

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Abstract—This paper presents an integrated gas monitoring system that combines stationary gas sensors with a mobile robotic agent for gas distribution mapping over a monitored area. The key idea is to leverage the mobility of the robotic agent to fill the gaps between stationary sensors to collect gas concentration samples to improve the gas mapping accuracy. An uncertainty-first robot path planning algorithm is presented to guide the robot toward under-sampled regions. The proposed approach is evaluated using the gas plume data collected from real-world wind tunnel experiments. Test results demonstrate that the integrated stationary–mobile sensing system improves the gas mapping accuracy compared to stationary sensors alone, and the proposed uncertainty-first planning strategy outperforms fixed robot path patterns, such as the lawn-mower robot path.

Index Terms—gas monitoring, robotics, sensor network

I. INTRODUCTION

Gas monitoring is a critical task in environmental monitoring. For instance, in Carbon Capture, Utilization, and Storage (CCUS) facilities, gas leak detection poses a critical operational challenge: undetected gas leaks, e.g., carbon dioxide (CO₂) leaks, can compromise environmental safety, reduce operational efficiency, and violate regulatory requirements [1], [2]. Conventional gas monitoring approaches [3]–[5] typically rely on a network of stationary sensors. However, deploying and maintaining a large number of sensors leads to substantial installation, calibration, and maintenance costs.

Recent advances in robotic technologies have enabled mobile robotic systems to play an increasingly important role in real-world environmental monitoring applications [6]–[8]. Mobile robots provide a flexible alternative by collecting spatially distributed measurements without requiring sensors at every location. When integrated with a sparse stationary sensor network, mobile robots can function as dynamic sensing agents that complement fixed sensors to improve spatial coverage and support timely leak detection. This integrated stationary–mobile sensing system has the potential to reduce the number of required stationary sensors while maintaining the overall gas monitoring accuracy.

The task of gas distribution mapping aims to estimate the spatial distribution of a target gas within an environment, often to identify hazardous regions or monitor long-term trends [9]. While traditional stationary sensor networks can generate

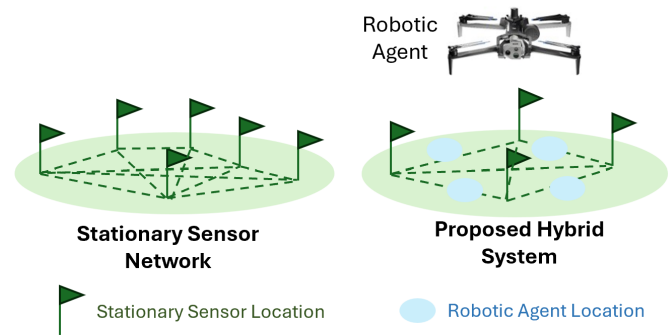


Fig. 1. The proposed system combines a mobile robot (e.g., a drone) with a stationary sensor network. The proposed system is designed to generate a gas concentration map of the monitored area.

high-resolution gas distribution maps, they often incur substantial costs related to sensor deployment, calibration, and maintenance, particularly in large-scale or complex environments [10]. To address these challenges, model-based plume mapping techniques have been developed to estimate the gas plume distribution from limited measurements. Methods such as hidden Markov process [11], Gaussian process [12], and the filament-based plume model [13] use point gas sensor measurements and wind information to estimate the gas plume distributions over the monitored areas.

In this work, we present an integrated system that comprises stationary sensors and a mobile robotic agent, as shown in Fig. 1. Compared to traditional stationary sensor networks, the addition of a mobile agent (e.g., a drone) enables the proposed system to fill spatial gaps between fixed sensor locations. To guide the robot's motion, we present an uncertainty-first path planning strategy that directs the robot toward under-sampled areas, including areas without stationary sensor coverage and regions that have not been visited by the robot. The proposed system is evaluated using the gas plume data collected from real-world wind tunnel experiments. Experimental results show that incorporating a mobile robot can improve the gas mapping accuracy compared to using stationary sensor alone. In addition, the proposed uncertainty-first search strategy achieves a higher gas mapping accuracy compared to fixed robot path patterns, such as the lawn-mower path.

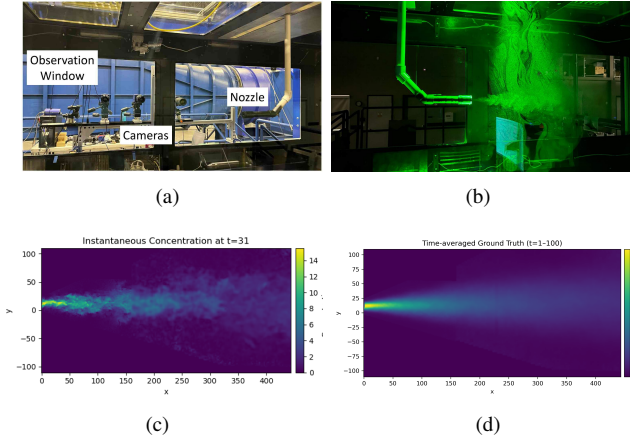


Fig. 2. Plume data acquisition. (a) The experimental setup; (b) The plume released into the wind tunnel; (c) The collected instantaneous plume field; (d) The collected time averaged plume field (1-100 sec).

II. METHODOLOGY

A. Plume Data Collection

Collecting real-world gas plume data is a challenging task due to the stochastic and turbulent nature of plume dispersion. In this work, we utilize the plume data collected from a controlled wind tunnel environment with a particle image velocimetry (PIV) system. The experimental setup, as illustrated in Fig. 2(a), consists of a wind tunnel with the observation window, a plume-releasing nozzle, and a high-resolution PIV system.

As shown in Fig. 2(b), the plume nozzle continuously releases plumes (mineral oil) into the wind tunnel, while the PIV system captures plume dynamics in the downstream region. Using this setup, 100 seconds of plume dispersion data, including plume speeds at x -axis and y -axis, was collected with 1 frame per second. To approximate the plumes concentrations, we assume that higher speeds correspond to denser plume areas. Thus, the plume concentration at a location can be calculated via:

$$c = \sqrt{u^2 + v^2}, \quad (1)$$

where c is the plume concentration, u and v are measured plume speeds at x -axis and y -axis, respectively. Representative results of plume fields are presented in Fig. 2(c) and Fig. 2(d): the instantaneous plume field and the time-averaged plume field (averaged over 100 seconds). In both figures, lighter colors correspond to higher plume concentrations.

B. Stationary Sensor Distribution

In this work, the objective of gas mapping is to generate a gas concentration map across a monitored area. A hybrid sensing system that combines measurements from a mobile robot and a set of stationary sensors is presented to construct this gas concentration map. The mobile robot is assumed to be equipped with a gas sensor that measures the gas concentration at its current location, while each stationary sensor provides measurements at its fixed position.

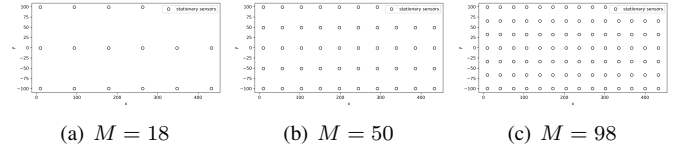


Fig. 3. Stationary sensor distribution. The number of stationary sensors is M , where each dot presents a stationary sensor location and measures gas concentrations at its location.

In the practical scenarios, the location of the gas source is typically unknown, thus stationary sensors cannot be deliberately positioned near the emission point. To address this uncertainty, we assume that M stationary sensors are evenly distributed throughout the monitored area to ensure the baseline spatial coverage.

Let the monitored area be denoted as Ω , which can be presented as:

$$\Omega = \{\mathbf{p} = (x, y) | x_{\min} \leq x \leq x_{\max}, y_{\min} \leq y \leq y_{\max}\}, \quad (2)$$

where $x_{\min}$, $x_{\max}$, $y_{\min}$, and $y_{\max}$ are limits of the monitored area. In this work, the monitored area corresponds to the plume field collected from wind tunnel experiments, with $x \in [0, 440]$ and $y \in [-110, 110]$, representing a rectangular region of size $5 \times 2.2 \text{ m}^2$.

The locations of stationary sensors are fixed and known, and are denoted by the set $\mathcal{S}_{\text{stat}} = \{\mathbf{p}_s^1, \mathbf{p}_s^2, \dots, \mathbf{p}_s^M\}$. To evenly distribute the stationary sensors, the number of sensors along the x -axis is set to be twice that along the y -axis. From sparse to dense network settings, Fig. 3 presents the proposed stationary sensor layouts for $M = 18$, $M = 50$, and $M = 98$, respectively.

C. Mobile Robot Path for Gas Mapping

To guide the robot motion, we introduce an uncertainty-first search strategy. The core idea is using numerical scores to quantify the spatial uncertainty of candidate locations based on their distances to existing measurements. Candidate locations close to stationary sensors or past robot positions are considered well explored and therefore assigned low scores, whereas locations far from any existing measurements are considered highly uncertain and will be assigned with high scores.

1) *Reachable Region and Planning Region*: During the survey, mobile robot locations are stored in a set $\mathcal{S}_{\text{mob}} = \{\mathbf{p}_r^1, \mathbf{p}_r^2, \dots, \mathbf{p}_r^k\}$, where k is the number of robot positions. We define the combined set of all existing measurement locations as:

$$\mathcal{S} = \mathcal{S}_{\text{mob}} \cup \mathcal{S}_{\text{stat}}. \quad (3)$$

These locations have already been sampled and should be avoided during the future robot exploration.

Based on the robot location $\mathbf{p}_r$, two spatial regions are defined (see Fig. 4):

- **Reachable Region $\mathcal{D}$** : locations where the robot can reach within one time step;
- **Planning Region $\mathcal{L}$** : locations that are considered during multi-step planning.

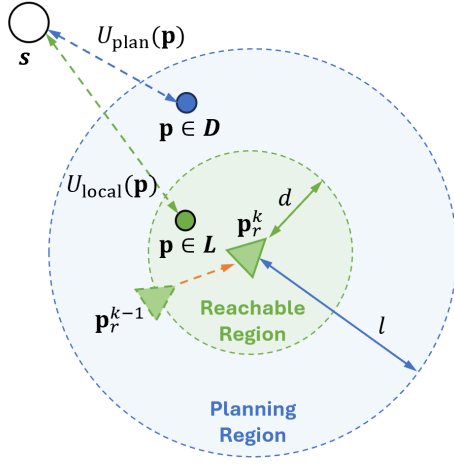


Fig. 4. The illustrative diagram for the proposed uncertainty-driven planning. The robot location is $\mathbf{p}_r$; reachable region, with the range of d , is $\mathbf{D}$; planning region, with the range of l , is $\mathbf{L}$; samples of existing measurements are in a set $\mathbf{S}$; uncertain score of a point in reachable region is $U_{\text{local}}(\mathbf{p})$ and $U_{\text{plan}}(\mathbf{p})$ for a point in the planning region.

These regions are defined as:

$$\begin{aligned} \mathbf{D} &= \{\mathbf{p} \mid \|\mathbf{p}_r - \mathbf{p}\|_2 \leq d\}, \\ \mathbf{L} &= \{\mathbf{p} \mid d < \|\mathbf{p}_r - \mathbf{p}\|_2 \leq l\}, \end{aligned} \quad (4)$$

where d denotes the maximal distance that the robot can move within one time step, and $l > d$ denotes the planning horizon distance. The Euclidean norm is defined as:

$$\|\mathbf{p}_r - \mathbf{p}\|_2 = \sqrt{(x_r - x)^2 + (y_r - y)^2}. \quad (5)$$

Notice that, the reachable region and the planning region is not overlapped.

2) *Uncertainty Score*: To plan the robot next location, each candidate location $\mathbf{p}$ is assigned with an uncertainty score, defined as the distance to the nearest sampled location (i.e., a location $\mathbf{s}$ in $\mathbf{S}$):

$$U(\mathbf{p}) = \min_{\mathbf{s} \in \mathbf{S}} \|\mathbf{p} - \mathbf{s}\|_2. \quad (6)$$

This score definition reflects the motivation to direct the robot toward unsampled areas: a candidate location $\mathbf{p}$ is well explored if at least one existing sampled location is nearby, and highly uncertain if it is far from all existing samples. According to this definition, we can find local uncertainty scores for all points over the reachable region:

$$U_{\text{local}}(\mathbf{p}) = \min_{\mathbf{s} \in \mathbf{S}} \|\mathbf{p} - \mathbf{s}\|_2, \quad \mathbf{p} \in \mathbf{D}, \quad (7)$$

and planning uncertainty scores for all points over the planning region:

$$U_{\text{plan}}(\mathbf{p}) = \min_{\mathbf{s} \in \mathbf{S}} \|\mathbf{p} - \mathbf{s}\|_2, \quad \mathbf{p} \in \mathbf{L}. \quad (8)$$

Based on uncertainty scores, we can determine the best next point within these two ranges via:

$$\begin{aligned} \mathbf{p}_{\text{opt,local}} &= \arg \max_{\mathbf{p} \in \mathbf{D}} U_{\text{local}}(\mathbf{p}), \\ \mathbf{p}_{\text{opt,plan}} &= \arg \max_{\mathbf{p} \in \mathbf{L}} U_{\text{plan}}(\mathbf{p}), \end{aligned} \quad (9)$$

which represents a short-term exploitation target and a long-term exploration target that may not be immediately reachable by the robot, respectively.

To balance these two targets, we first transfer the long-term target within the reachable region via calculating the distance between all points within the reachable region to the long-term target: $d(\mathbf{p}) = \|\mathbf{p} - \mathbf{p}_{\text{opt,plan}}\|_2$, $\mathbf{p} \in \mathbf{D}$. Then, we define a combined score as

$$q(\mathbf{p}) = \lambda \hat{U}_{\text{local}}(\mathbf{p}) - (1 - \lambda) \hat{d}(\mathbf{p}), \quad \mathbf{p} \in \mathbf{D}, \quad (10)$$

where $\hat{U}_{\text{local}}(\mathbf{p})$ and $\hat{d}(\mathbf{p})$ denote the normalized local uncertainty and the normalized distance, respectively. Here, min-max normalization is employed. $\lambda \in [0, 1]$ is a tuning parameter: when $\lambda = 1$, the robot performs purely local uncertainty-driven search; when $\lambda = 0$, the robot moves greedily toward the long-term target.

Finally, the robot selects the next sampling location as:

$$\mathbf{p}_{\text{opt}} = \arg \max_{\mathbf{p} \in \mathbf{D}} q(\mathbf{p}). \quad (11)$$

This will make sure that the point selected will be reachable for the robot.

D. Interpolate the Gas Concentration Map

After obtaining concentration measurements and their corresponding locations from both the stationary sensors and the mobile robot, a gas concentration map over the monitored area is reconstructed using linear interpolation. This interpolation estimates concentration values at locations that are not directly sampled by either the stationary sensors or the mobile robot.

At each survey time step, the stationary sensor network provides M concentration measurements, while the mobile robot collects n measurements along its trajectory. Due to the high flexibility of mobile robots (e.g., drones), it is assumed that the mobile robot is capable of moving n steps within a single time step, thereby collecting concentration measurements at n locations. As a result, a total of $M + n$ measurement locations and their associated concentration values are available for interpolation at each time step. In this work, the linear interpolation is performed using the Python SciPy library [14].

III. EXPERIMENTS AND RESULTS

A. Experiment Setup

The gas plume fields collected from real-world wind tunnel experiments are used as the monitored environment in this work, where the spatial domain is defined as $x \in [0, 440]$ and $y \in [-110, 110]$. A total of M stationary sensors are deployed within the environment, with $M = \{18, 50, 98\}$, representing scenarios ranging from sparse to dense sensor networks. Fig 3 presents the distribution of these stationary sensors.

For the mobile robot, we fixed the robot initial location at the lower-right corner of the monitored area, i.e., (431, -98), with the initial heading of 90 degrees. The robot is assumed to be equipped with a gas sensor capable of measuring the gas concentration at its current location. The maximal step length of the robot in a single movement is set to $d = 15$, and the

TABLE I
RMSE FOR DIFFERENT $l = z \cdot d$ AND λ WITH $M = 18$.

$z \setminus \lambda$	0.1	0.2	0.5	0.8
3	1.0906	1.0987	1.0942	1.1004
5	1.0831	1.0913	1.0911	1.1039
10	1.0940	1.0824	1.0930	1.0893

TABLE II
RMSE FOR DIFFERENT $l = z \cdot d$ AND λ WITH $M = 50$.

$z \setminus \lambda$	0.1	0.2	0.5	0.8
3	0.9962	0.9912	0.9911	0.9934
5	0.9895	0.9971	0.9869	0.9941
10	0.9853	0.9833	0.9875	0.9917

robot can move $n = 5$ movement steps within a single time step.

In addition to the proposed uncertainty-first planning strategy, a lawn-mower coverage path is employed as a baseline for comparison. In this baseline approach, the robot follows a predetermined sweeping trajectory to survey the entire monitored area, independent of the stationary sensor locations. A buffer region is defined near the boundaries of the monitored area to prevent collisions with the environment boundaries. The robot starts from the lower-right corner and moves upward toward the top boundary. Upon reaching either the top or bottom boundary, the robot shifts leftward by a short distance and reverses its heading to continue the sweeping motion. The trajectory terminates when the robot reaches the left boundary of the monitored area.

To evaluate gas mapping performance, a time-averaged gas concentration map is computed and compared against the ground truth concentration map obtained from the wind tunnel data. The mapping accuracy is measured via the root mean square error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega|} \sum_{(x,y) \in \Omega} (\hat{c}(x,y) - c(x,y))^2}, \quad (12)$$

where $\hat{c}$ denotes the estimated gas concentration obtained from the interpolated gas concentration map (using either stationary sensors alone or the integrated stationary-mobile sensing system), and $c(x,y)$ represents the ground truth gas concentration. All concentrations considered in this evaluation are time-averaged values.

B. Evaluation on λ and l

We first evaluate the effects of weighting parameter λ and the planning horizon l on gas mapping accuracy. Table I, II, and III report the RMSE values of the proposed uncertainty-first search strategy under different combinations of λ and l for varying numbers of stationary sensors. The planning horizon is defined as $l = z \cdot d$, where $z = \{3, 5, 10\}$ denotes the ratio between the planning horizon and the robot's maximum step length d . The weighting parameter is selected from $\lambda = \{0.1, 0.2, 0.5, 0.8\}$. In this evaluation, the number of stationary sensor is set as $M = \{18, 50, 98\}$.

TABLE III
RMSE FOR DIFFERENT $l = z \cdot d$ AND λ WITH $M = 98$.

$z \setminus \lambda$	0.1	0.2	0.5	0.8
3	0.9381	0.9447	0.9462	0.9395
5	0.9428	0.9453	0.9361	0.9360
10	0.9326	0.9414	0.9382	0.9364

As shown in as shown in these tables, the proposed integrated stationary-mobile sensing system achieves the lowest RMSE value when $\lambda = 0.2$ and $l/d = 10$ in sparse sensor network scenarios (i.e., $M = 18$ and 50), indicating the smallest gap between the estimated gas concentration map and the ground truth. In dense sensor network case ($M = 98$), the combination of $\lambda = 0.1$ and $l/d = 10$ provides the best mapping accuracy with the lowest RMSE.

C. Comparison with the Baseline and Stationary-only System

In this section, the best combination of λ and l , obtained from the previous section, is employed in the robot path planning. Gas mapping results of the proposed uncertainty-first algorithm are compared with: (i) a stationary sensor only system and (ii) the sensor and mobile robot integrated system where the mobile robot moves in a fixed lawn mower path.

Fig. 5 presents the time-averaged gas concentration maps produced by these three approaches, while Table IV reports their corresponding RMSE values under different numbers of stationary sensors. It should be noted that the lawn-mower path does not span the full 100-second monitoring duration; instead, it completes its coverage in 70 seconds. Consequently, both the concentration maps and RMSE values for the lawn-mower baseline are computed using time-averaged data over 70 seconds, whereas the other methods are averaged over 100 seconds.

From the results, it is observed that increasing the number of stationary sensors consistently reduces the mapping error for all methods, as the stationary sensor network alone provides denser spatial coverage. Moreover, incorporating a mobile robot leads to lower RMSE values across all sensor densities, demonstrating the benefit of mobile sensing in improving gas mapping accuracy.

When comparing the two mobile robot planning strategies, the proposed uncertainty-first search achieves the lowest RMSE in all sensor density scenarios, outperforming the lawn-mower baseline. This result suggests that the proposed robot planning algorithm, which actively directs the robot toward under-sampled regions, can improve the gas mapping accuracy, compared to the lawn-mower baseline.

IV. CONCLUSION

In this work, we present an integrated gas monitoring system that combines stationary sensors with a mobile robot to generate a gas concentration map over a monitored area. To guide the robot's motion, we propose an uncertainty-first search strategy that directs the robot toward regions not covered by stationary sensors and not previously visited. The proposed approach is evaluated using gas plume data collected from

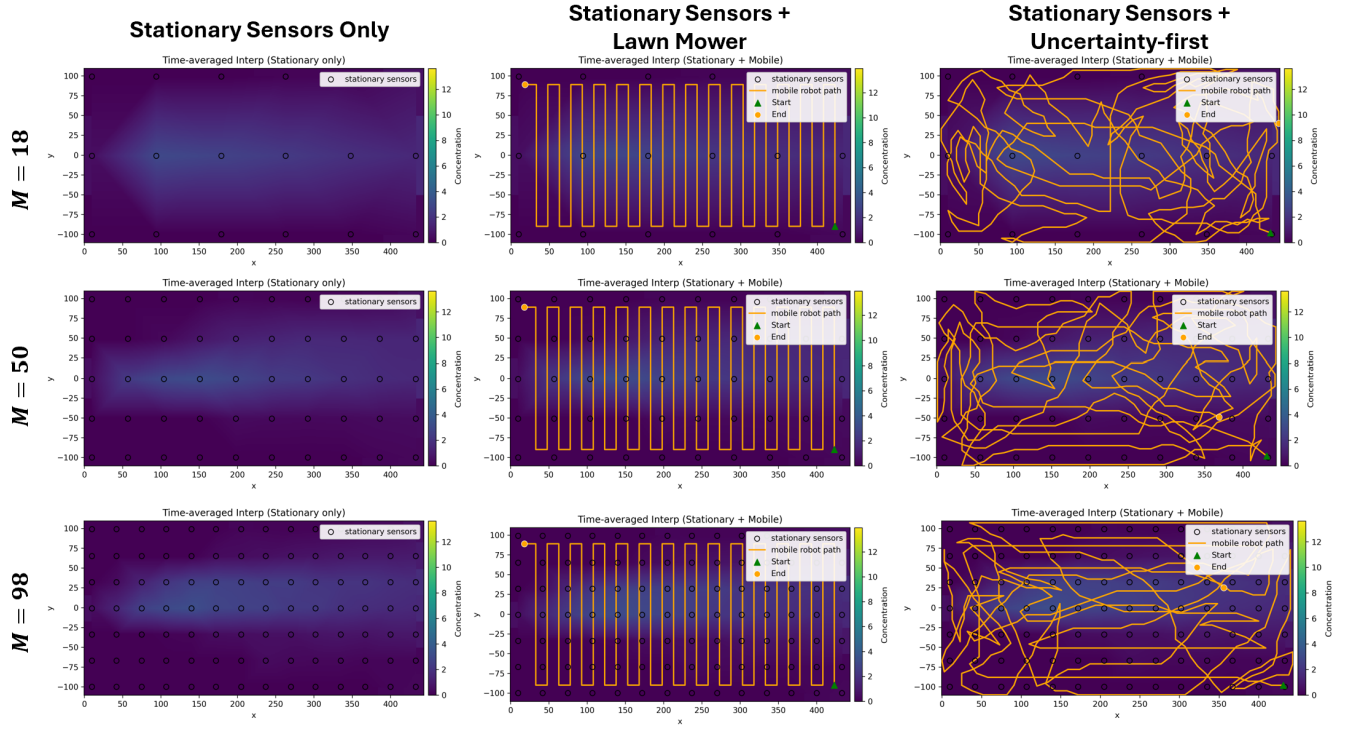


Fig. 5. Time-averaged gas concentration maps obtained using stationary sensors only (first column), the lawn-mower baseline (second column), and the proposed uncertainty-first method (third column). Rows correspond to different numbers of stationary sensors: $M = 18$ (first row), $M = 50$ (second row), and $M = 98$ (third row). Stationary sensor locations are represented as black circles; mobile robot initial location is a green triangle (lower-right corner); and the robot end location is a yellow dot.

TABLE IV
RMSE OF THE STATIONARY SENSOR ONLY, LAWN-MOWER, AND THE PROPOSED UNCERTAINTY-FIRST METHOD

	Stationary Sensors Only	Stationary Sensors + Lawn Mower	Stationary Sensors + Uncertainty-First
$M = 18$	1.1219	1.0945	1.0824
$M = 50$	1.0092	0.9963	0.9833
$M = 98$	0.9530	0.9364	0.9326

real-world wind tunnel experiments, where the RMSE between the reconstructed gas concentration maps and the ground truth is used as the performance metric. Experimental results demonstrate that integrating a mobile robot with stationary sensors can improve the gas mapping accuracy compared to using stationary sensors alone. Moreover, the proposed uncertainty-first search strategy achieves the highest mapping accuracy among all compared methods.

Future works include implementing the proposed algorithm into a real-world testing field. Also, besides the uncertainty scores, other factors could also be considered into the robot path planning. For instance, in real-world settings, we can encourage the robot to move toward downwind areas where gas leaks are more likely to be detected. In addition, we can also integrated the gas plume tracing algorithms to the robotic agent to enable it trace emitted gas plumes back to the origin (i.e., the leak location).

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REFERENCES

- [1] G. Cohen, C. Loisy, C. Laveuf, O. Le Roux, P. Delaplace, C. Magnier, V. Rouchon, B. Garcia, and A. Cerepi, "The co2-vadose project: Experimental study and modelling of co2 induced leakage and tracers associated in the carbonate vadose zone," *International Journal of Greenhouse Gas Control*, vol. 14, pp. 128–140, 2013.
- [2] E. H. Keating, J. A. Hakala, H. Viswanathan, J. W. Carey, R. Pawar, G. D. Guthrie, and J. Fessenden-Rahn, "Co2 leakage impacts on shallow groundwater: Field-scale reactive-transport simulations informed by observations at a natural analog site," *Applied geochemistry*, vol. 30, pp. 136–147, 2013.
- [3] L. B. Frederickson, R. Sidaraviciute, J. A. Schmidt, O. Hertel, and M. S. Johnson, "Are dense networks of low-cost nodes really useful for monitoring air pollution? a case study in staffordshire," *Atmospheric Chemistry and Physics*, vol. 22, no. 21, pp. 13 949–13 965, 2022.
- [4] W. Tsujita, A. Yoshino, H. Ishida, and T. Morizumi, "Gas sensor network for air-pollution monitoring," *Sensors and Actuators B: Chemical*, vol. 110, no. 2, pp. 304–311, 2005.
- [5] A. Ortiz Perez, B. Bierter, L. Scholz, J. Wöllenstein, and S. Palzer, "A wireless gas sensor network to monitor indoor environmental quality in schools," *Sensors*, vol. 18, no. 12, p. 4345, 2018.
- [6] H. Ishida, Y. Wada, and H. Matsukura, "Chemical sensing in robotic applications: A review," *IEEE Sensors Journal*, vol. 12, no. 11, pp. 3163–3173, 2012.
- [7] X.-x. Chen and J. Huang, "Odor source localization algorithms on mobile robots: A review and future outlook," *Robotics and Autonomous Systems*, vol. 112, pp. 123–136, 2019.

- [8] T. Jing, Q.-H. Meng, and H. Ishida, "Recent progress and trend of robot odor source localization," *IEEE Transactions on Electrical and Electronic Engineering*, vol. 16, no. 7, pp. 938–953, 2021.
- [9] N. P. Winkler, O. Kotlyar, E. Schaffernicht, H. Matsukura, H. Ishida, P. P. Neumann, and A. J. Lilienthal, "Super-resolution for gas distribution mapping," *Sensors and Actuators B: Chemical*, vol. 419, p. 136267, 2024.
- [10] H. Mei, P. Wei, Y. Wang, M. A. Ghadikolaie, N. K. Gali, and Z. Ning, "In-situ baseline calibration approach for enhanced data quality of large-scale air sensor monitoring networks," *npj Climate and Atmospheric Science*, vol. 8, no. 1, p. 296, 2025.
- [11] J. A. Farrell, S. Pang, and W. Li, "Plume mapping via hidden markov methods," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 33, no. 6, pp. 850–863, 2003.
- [12] M. Vergassola, E. Villermaux, and B. I. Shraiman, "'infotaxis' as a strategy for searching without gradients," *Nature*, vol. 445, no. 7126, pp. 406–409, 2007.
- [13] J. A. Farrell, J. Murlis, X. Long, W. Li, and R. T. Cardé, "Filament-based atmospheric dispersion model to achieve short time-scale structure of odor plumes," *Environmental fluid mechanics*, vol. 2, no. 1, pp. 143–169, 2002.
- [14] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors, "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.